Experiment 7: Bag of Words and TF-IDF Representation

A. Bag of Words (BoW)

```
from sklearn.feature_extraction.text import CountVectorizer
import pandas as pd

corpus = [
    "I loved the movie, it was it fantastic!",
    "The movie was okay, but not great.",
    "I hated the movie, it was terrible."
]

# Convert documents into word-count vectors
vectorizer = CountVectorizer()
document_matrix = vectorizer.fit_transform(corpus)

# Vocabulary
features = vectorizer.get_feature_names_out()

print("Vocabulary:")
print(features)

print("\nDocument-Term Matrix:")
print(document_matrix.toarray())

# Total word frequencies across the corpus
word_frequencies = document_matrix.toarray().sum(axis=0)

frequency_table = pd.DataFrame({
    "Word": features,
    "Frequency": word_frequencies
}).sort_values(
    by=["Frequency", "Word"],
    ascending=[False, True]
)

print("\nWord Frequencies:")
print(frequency_table.to_string(index=False))
```

```
Vocabulary:
['but' 'fantastic' 'great' 'hated' 'it' 'loved' 'movie' 'not' 'okay'
 'terrible' 'the' 'was']

Document-Term Matrix:
[[0 1 0 0 2 1 1 0 0 0 1 1]
 [1 0 1 0 0 0 1 1 1 0 1 1]
 [0 0 0 1 1 0 1 0 0 1 1 1]]
```

```
Word Frequencies:
     Word  Frequency
       it          3
    movie          3
      the          3
      was          3
      but          1
fantastic          1
    great          1
    hated          1
    loved          1
      not          1
     okay          1
 terrible          1
```

B. TF-IDF Representation

```
from sklearn.feature_extraction.text import TfidfVectorizer
import pandas as pd

corpus = [
    "NLP is fun",
    "NLP helps computers understand language",
    "Language is power"
]

# Create and fit the TF-IDF vectorizer
vectorizer = TfidfVectorizer()
tfidf_matrix = vectorizer.fit_transform(corpus)

features = vectorizer.get_feature_names_out()

print("Feature Names:")
print(features)

print("\nTF-IDF Matrix:")
print(tfidf_matrix.toarray())

# Display as a labeled table
tfidf_table = pd.DataFrame(
    tfidf_matrix.toarray(),
    columns=features,
    index=[f"Document {i}" for i in range(1, len(corpus) + 1)]
)

print("\nLabeled TF-IDF Table:")
print(tfidf_table.round(4))
```

```
Feature Names:
['computers' 'fun' 'helps' 'is' 'language' 'nlp' 'power' 'understand']
```

```
TF-IDF Matrix:
[[0.         0.68091856 0.         0.51785612 0.         0.51785612
  0.         0.        ]
 [0.49047908 0.         0.49047908 0.         0.37302199 0.37302199
  0.         0.49047908]
 [0.         0.         0.         0.51785612 0.51785612 0.
  0.68091856 0.        ]]

Labeled TF-IDF Table:
            computers     fun   helps      is  language     nlp   power  \
Document 1     0.0000  0.6809  0.0000  0.5179    0.0000  0.5179  0.0000
Document 2     0.4905  0.0000  0.4905  0.0000    0.3730  0.3730  0.0000
Document 3     0.0000  0.0000  0.0000  0.5179    0.5179  0.0000  0.6809

            understand
Document 1      0.0000
Document 2      0.4905
Document 3      0.0000
```